AlphaDraft
ARCHITECTURE & ENGINEERING HANDOVER

OFFICIAL MEMO

TO
Executive Leadership & Engineering Management

FROM
AlphaDraft Core Engineering & Architecture Team

DATE
October 10, 2026

SUBJECT
Full Decoupling Migration, Security Hardening & Handover

1. Executive Summary

This handover note confirms the successful architectural audit, stabilization, security hardening, and documentation of the **AlphaDraft** web platform (`alphadrafts-new`). The application has fully decoupled from legacy Firebase SDKs and now operates on an independent, enterprise cloud-native infrastructure.

> **Production Stability Verification**
> The codebase has achieved **zero build and lint defects**. Both `tsc --noEmit` and `next lint` execute with 0 errors and 0 warnings, verified on the latest production commit (`b3dcf52`) on branch `main`.

2. Architectural Foundations

- **Decoupled Cloud Engine:** Cloud Run REST backend backed by PostgreSQL (Drizzle ORM) and Cloudflare R2 object storage.
- **Bank-Grade Cookie Authentication:** httpOnly, SameSite, Secure JWT tokens with automated background rotation and double-submit CSRF headers.
- **Design System Tokens:** Material 3 design system with dynamic light/dark theming, WCAG 2.1 motion preference support, and Tailwind CSS v4 tokens.
- **Advanced Document Editing:** TipTap PM v3 rich text editor featuring debounced 1.5s background auto-saving, rollback snapshots, and inline AI highlights.

3. Quality & Security Hardening Scorecard

AREA	STATUS	RESULT & IMPACT
TypeScript Safety	0 ERRORS	Full type coverage across React 19 contexts, API responses, and TipTap models.
ESLint Standards	0 WARNINGS	ESLint Flat config updated with build artifact exclusions and clean dependencies.
AI Proxy Security	HARDENED	Secured `/api/v1/sapling/*` behind cryptographic session gates to prevent quota leakage.
CSRF Defense	ENFORCED	Double-submit cookie validation active on all state-mutating requests (POST/PUT/DELETE).

4. Central Documentation Repository

ARCHITECTURE.md
System diagrams, Cloud Run proxy flows, and storage mechanics.

DATABASE_SCHEMA.md
PostgreSQL ERD, table dictionaries, relations, and index policies.

APP_FLOW.md
Authentication lifecycles, project wizard stages, and auto-save state.

TRD.md
Technical Requirements, NFRs, SLA targets, and API contracts.

5. High-Priority Engineering Recommendations for Next Sprint

Action Items for Next Release Cycle:

1. **Consolidate Date Formatting:** Standardize on `date-fns` across `formatDates.ts` to retire legacy date helper functions.
2. **Mailing List Hardening:** Integrate rate limiting and Turnstile captcha on `/api/v1/mail-list` to safeguard against automated form submissions.
3. **Dependency Optimization:** Prune legacy packages (such as `jquery` and secondary docx converters) to further streamline client bundle sizes.